

Full Stack Development with MERN Project

Documentation

1. Introduction

- Project Title: Book A Doctor
- Team Members:

Name	Role
Srimai Gopala	Frontend Development, Backend Development, Database Management, Testing

2. Project Overview

Purpose :

Book-A-Doctor is a healthcare appointment management platform developed to simplify the process of finding doctors and scheduling medical consultations online. Traditional appointment booking methods often require patients to visit hospitals physically or make phone calls, which can be time-consuming and inconvenient. This system provides a centralized platform where patients can search for doctors, book appointments, and track appointment status efficiently.

The platform also provides dedicated interfaces for doctors and administrators. Doctors can manage appointment requests and update their professional profiles, while administrators can verify doctors and monitor overall platform activities. The system ensures secure communication and role-based access using JWT authentication.

Features :

- Patient Registration and Login
- Doctor Registration and Onboarding
- Admin Verification of Doctors
- Doctor Search by Specialization
- Appointment Booking System
- Appointment Approval/Rejection Workflow
- Patient Dashboard
- Doctor Dashboard
- Admin Dashboard
- Secure JWT Authentication
- Role-Based Access Control

3. Architecture

Frontend Architecture :

The frontend of Book-A-Doctor is developed by React.js and Vite. React components are responsible for creating various UI components like login pages, dashboards, appointment cards, etc. Axios allows communication between frontend and backend APIs. The frontend is designed separately for Patients, Doctors, and Administrators. Each user can only visit web pages according to their designated roles via protected routes.

Backend Architecture :

Node.js and Express.js frameworks are used for the development of the backend. REST APIs take care of authentication, onboarding of doctors, appointments, etc. While controllers handle the business logic, routes handle the incoming requests. The implementation of JWT middleware helps validate user sessions on the server side.

Database Architecture :

The database chosen is MongoDB Atlas and serves as a cloud database. It holds data related to the system including user accounts, doctor profiles, appointment details, etc.

System flow:

1. The Patient and Doctor have to register themselves by providing their email address and password.
2. The Doctor has to fill up his/her information like specialization, experience, and consultation fees.
3. The Admin confirms registration of Doctors; if approved, they can register else will be rejected.
4. The Patient will find doctors through search and book appointments with verified Doctors.
5. The Doctor either approves or rejects the appointment requested by the Patient.
6. The Patient sees the status of the Appointment (Pending, Approved, Rejected, or Cancelled).

4. Setup Instructions

Prerequisites

Before running the application, ensure the following software is installed:

- Node.js v18 or above
- npm
- MongoDB Atlas Account
- Git
- VS Code

Installation Steps:

Clone Repository:

```
git clone <repository-url>
```

Install Frontend Dependencies:

```
cd client  
npm install
```

Install Backend Dependencies:

```
cd server  
npm install
```

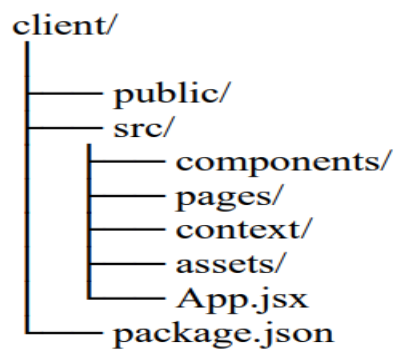
Environment Variables:

Create .env

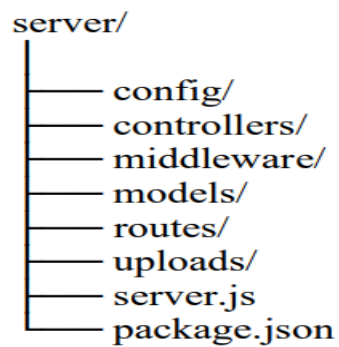
```
MONGO_URI=your_mongodb_connection_string  
JWT_SECRET=your_secret_key  
PORT=5000
```

5. Folder Structure

- **Client:**



- **Server:**



6. Running the Application

Start Backend

```
cd server  
npm start
```

Backend URL:

<http://localhost:5000>

Start Frontend

```
cd client  
npm run dev
```

Frontend URL:

<http://localhost:3000>

7. API Documentation

Authentication APIs

Register User

POST /api/auth/register

Login User

POST /api/auth/login

Doctor APIs

Submit Doctor Profile

POST /api/doctors/onboarding

Get Doctors

GET /api/doctors

Appointment APIs

Create Appointment

POST /api/appointments

View Appointments

GET /api/appointments

Update Appointment Status

PUT /api/appointments/:id

8. Authentication

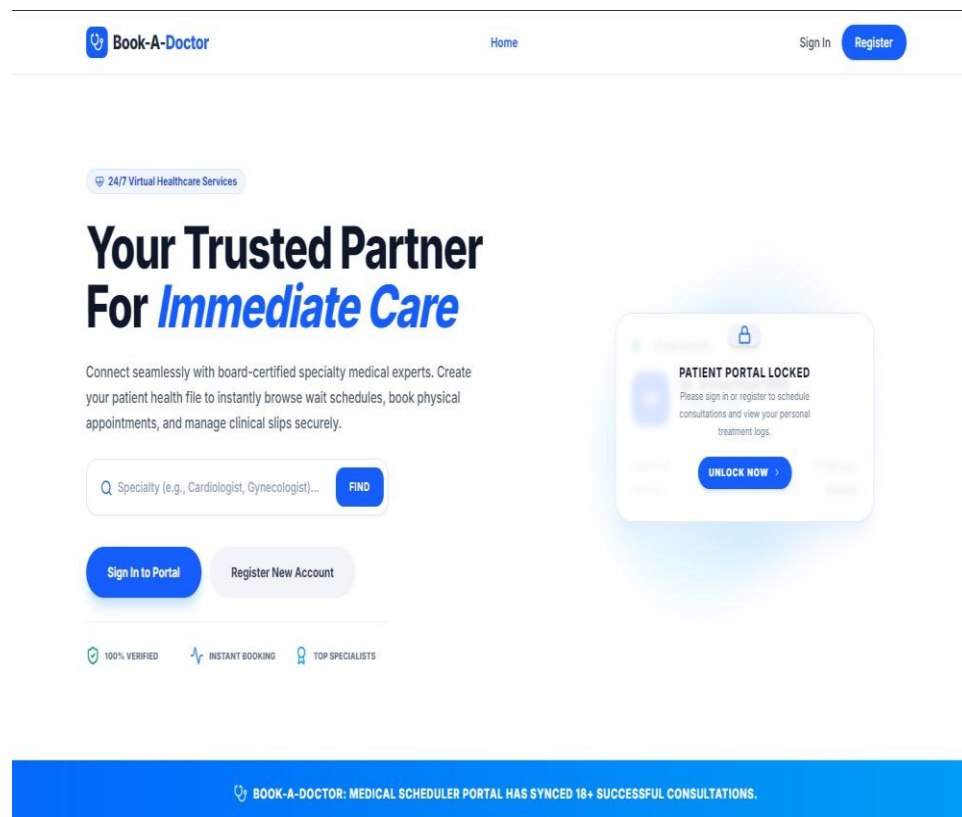
The Book-A-Doctor application uses JWT (JSON Web Token) based authentication to protect user sessions. Once a user logs in successfully, a JWT token will be issued and stored at the client-side. It will be sent along with all future requests for authentication purposes.

Some security practices employed include:

- Password hashing with bcrypt
- JWT Authentication
- Protected routes
- Role-based access control
- Secure API validation

9. User Interface

Home Page :



Doctor Registration :



Create Your Account

Join as a Doctor — submit your details for admin verification

SELECT REGISTRATION TYPE

PATIENT

DOCTOR

FULL NAME

Rosy

EMAIL ADDRESS

Rosy@gmail.com

PASSWORD

.....

CREATE ACCOUNT →

Already have a clinical account? [Sign in](#)

Doctor Onboarding :

 Book-A-Doctor

Home Dashboard

Rosy
DOCTOR Logout



Complete Your Profile

Welcome, Dr. Rosy! Please provide your clinical details to complete your registration.

CLINICAL SPECIALTY

Gynecologist

CLINICAL EXPERIENCE (YEARS)

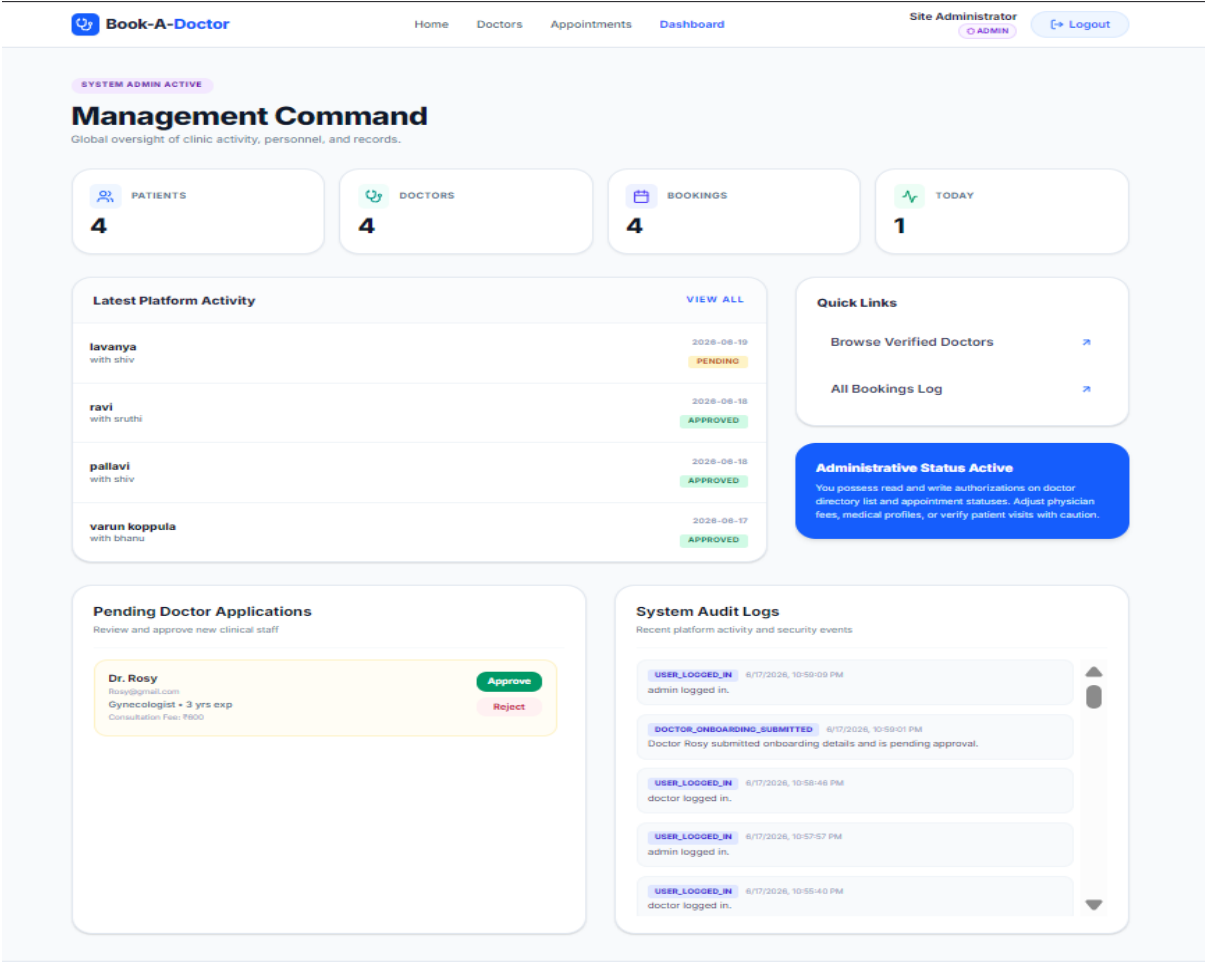
3

CONSULTATION FEES (₹)

₹ 600

SUBMIT FOR VERIFICATION

Admin Dashboard:



Doctors List :

 **Book-A-Doctor**

HomeDoctorsAppointmentsDashboard

ravi
PATIENTLogout

Our Registered Physicians

Browse, search, and schedule consultations with board-verified doctors.

QUICK FILTER:

ALLCARDIOLOGISTGENERAL PHYSICIANNEUROLOGISTGYNECOLOGIST

B

ONLINE • 10 yrs exp

bhanu
Cardiologist

FEE
₹1000

CHECK SCHEDULE

S

ONLINE • 4 yrs exp

shiv
Cardiologist

FEE
₹500

CHECK SCHEDULE

S

ONLINE • 3 yrs exp

sruthi
General Physician

FEE
₹200

CHECK SCHEDULE

S

ONLINE • 5 yrs exp

srilatha
Neurologist

FEE
₹600

CHECK SCHEDULE

R

ONLINE • 3 yrs exp

Rosy
Gynecologist

FEE
₹600

CHECK SCHEDULE

Book Appointment Page :

 **Book-A-Doctor**

HomeDoctorsAppointmentsDashboard

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Schedule Doctor Consultation

Book an appointment for on-duty diagnostic consulting.

SELECT SPECIALIST CLINICIAN

 Rosy - Gynecologist (Fee: ₹600)

✓ Lock set on: Rosy

DR

Rosy
GYNECOLOGIST • 3 Years Active Practice • Fee: ₹600

PREFERRED CONSULTATION DATE

19-06-2026

SELECT AVAILABLE TIME SLOT

09:00 AM

10:00 AM

11:00 AM

12:00 PM

02:00 PM

03:00 PM

04:00 PM

05:00 PM

REASON FOR VISIT

check up

This is shared with the doctor to help them review your request.

Proceed with Booking →

10. Testing

Testing was performed to ensure system reliability and functionality.

Functional Testing

Verified all modules work according to requirements.

Authentication Testing

Verified login, registration, token validation, and role-based access.

Appointment Testing

Verified booking, approval, rejection, and cancellation workflows.

UI Testing

Verified responsiveness and usability across different screen sizes.

11. Screenshots or Demo

Project Demo Link :

<https://drive.google.com/file/d/1dwDObWU71Zg6lupEyDUMBiVa0dRZl17h/view?usp=drivesdk>

12. Known Issues

- Email notifications are not integrated.
- Online payment system is not implemented.
- Video consultation functionality is unavailable.
- Mobile optimization can be improved further.
- Appointment reminder service is not implemented.

13. Future Enhancements

Future improvements planned for Book-A-Doctor include:

- Online Video Consultation
- AI-Based Doctor Recommendation System
- Digital Prescription Management
- Medical Report Upload and Storage
- Payment Gateway Integration
- SMS and Email Notifications
- Appointment Reminder System
- Multi-Language Support
- Hospital Management Integration
- Telemedicine Support

